DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 15: Risk-Neutral Probabilities

A Cleaner Pricing Formula · Why Real-World Probabilities Vanish

September 2, 2026

Today's Agenda

1. Repackaging the replication formula

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2. Why q is not a real probability
3. The tree, two ways
4. The no-arbitrage condition $d < 1 + r < u$

Learning Objectives

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- Price via risk-neutral valuation; confirm it matches replication.
- Explain why q doesn't depend on real-world probabilities.
- State and justify $d < 1 + r < u$.

The Risk-Neutral Valuation Formula

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Risk-Neutral Valuation

$$C_0 = \frac{1}{1+r}\Big[qC_u + (1-q)C_d\Big], \qquad q = \frac{(1+r)-d}{u-d}$$

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Remark

Whenever $d < 1 + r < u$: $q \in (0, 1)$.

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Remark

Whenever $d < 1 + r < u$: $q \in (0, 1)$. $qC_u + (1 - q)C_d$ is an expected value under q, discounted at r – an ordinary present-value calculation from Week 1.

Re-Pricing, the Fast Way

S_0=100, u=1.2, d=0.9, r=5% (C_u=15, C_d=0)

$$q = \frac{1.05 - 0.9}{1.2 - 0.9} = 0.5$$

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Identical to Lecture 14's replication answer.

A Fictional Coin, Useful Anyway

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Does Belief Change the Price?

Real-world $p = 0.7$ (optimists) vs. $p = 0.3$ (pessimists). Does C_0 change? **Not at all** – the formula for q never mentions p. We price in a fictional risk-neutral universe – not because it's true, but because it gives the correct arbitrage-free price.

Why p Never Appears

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The replicating portfolio used only S_0, S_u, S_d, r – observable by everyone, regardless of belief. Since C_0 equals that portfolio's cost, and the portfolio never used p, neither can C_0.

Two investors who disagree completely about p must still agree on C_0.

Checking q's Defining Property

$$\mathbb{E}^Q[S_1] = 0.5(120) + 0.5(90) = 105 = S_0(1 + r)$$

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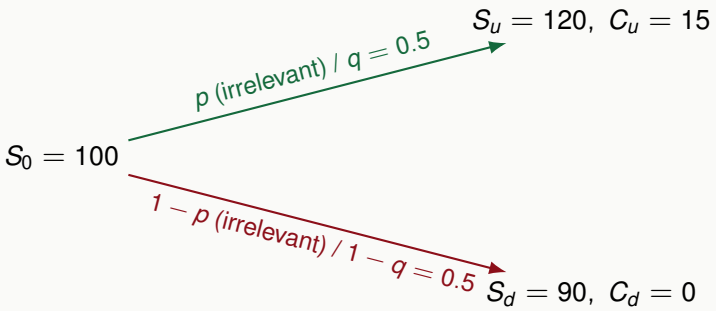
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Real investors aren't risk-neutral (Week 2) – q is a pricing device, not a belief.

The Tree, Two Ways



Only q ever enters the price.

Why We Need $d < 1 + r < u$

Key Takeaway

For $q \in (0, 1)$, and for the model to be arbitrage-free at all: $d < 1 + r < u$.

What Goes Wrong If This Fails

Suppose $1 + r \geq u$: e.g., r=25%, $1 + r$=1.25 $\geq u$=1.2

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q's formula would give $1.167 > 1$ *– nonsense, warning you the model is broken.*

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$d < 1 + r < u$: the risk-free return must sit strictly between the stock's worst and best outcomes. If r beat the stock's best case, no one would ever bear stock risk. If r were worse than the stock's worst case, the stock would dominate the bond always. Either way – not a sane market.

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- Risk-neutral valuation: $C_0 = \frac{1}{1+r}[qC_u + (1-q)C_d]$, same answer as replication.
- q never depends on real-world p – a pricing device, not a belief.
- Under q, every asset grows at exactly r.
- No-arbitrage requires $d < 1 + r < u$.

Check Your Understanding

Discuss (2 minutes)

1. $S_0=50$, $u=1.1$, $d=0.85$, $r=4\%$, put $K=52$. Find q; re-price via risk-neutral valuation.

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1. $S_0=50$, $u=1.1$, $d=0.85$, $r=4\%$, put $K=52$. Find q; re-price via risk-neutral valuation.
2. Why must two investors who disagree about p still agree on the option's price?
3. $u=1.15$, $d=0.95$. Find the range of r keeping the model arbitrage-free.

Looking Ahead

Module 3's one-period toolkit is complete: no-arbitrage, replication, risk-neutral pricing.

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This Week's Lab: The Arbitrage Hunter

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Week 6: from one coin flip to a full multi-period tree.